Week 9

Control

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Two loops, opposite ends of the vehicle

	Inner	Outer
Job	Hold a commanded steering angle	Follow a path
Runs on	Teensy 4.1	Laptop
Rate	200 Hz	~20 Hz
Method	PID	Pure pursuit

They have almost nothing in common except the word **control**.

1. PID

What each term buys, and costs

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \frac{de}{dt}$$

Term	Buys	Costs
P	Speed of response	Steady-state offset; oscillation
I	Kills steady-state offset	Windup, overshoot, sluggishness
D	Damping	**Amplifies sensor noise**

D on a noisy encoder cooks a gearmotor

Differentiating a noisy signal amplifies the noise.

On a mechanical steering column that is not just **seen** — it is **actuated**.
Audible chatter, and real current draw.

- Filter the derivative
- Or start at PD with a small D
- Or use derivative-on-**measurement**, so a setpoint step does not produce a spike

Where the loop lives

Not in ROS. Two reasons.

Timing. Linux cannot promise 200 Hz. A loop that **usually** makes its deadline oscillates unpredictably under load.

Survival. If the USB link drops, a loop running in ROS is **gone**.
A loop on the Teensy is still there — and can centre the wheels and de-energise, which is what failsafe row 1 requires.

If the loop cannot hold ≤ 1° steady-state with no sustained oscillation at Stage 1, adopt the daughterboard rather than continuing to tune.

E4 — written **before** anyone started tuning.

Why pre-registration matters

The decision to stop tuning is otherwise made by someone who is **three days into tuning** and emotionally invested in it.

Electrical track: ≤ 1° is your number, and Merge 1 is where it fires.

Invoking E4 costs you nothing here — the record already calls it the planned action.

2. Path tracking

A different problem entirely

There is no setpoint in the same sense.

There is a **path**, and the vehicle is somewhere near it.

Pure pursuit

Pick a point on the path a lookahead distance L_d ahead.

Compute the arc from here to there.

$$\kappa = \frac{2 \sin \alpha}{L_d} \qquad \delta = \arctan(\kappa L)$$

- No error term
- No gains
- No integrator

Pure geometry — which is why it cannot ask for a manoeuvre the car cannot perform.

Lookahead is the one real knob

L_d	**Behaviour**
Too short	Tracks tightly, oscillates, chases every wiggle
Too long	Smooth and stable, but **cuts corners**

What "Regulated" adds

- **Curvature regulation** — slow down in tight turns
- **Proximity regulation** — slow down near obstacles
- **Velocity-scaled lookahead** — longer when fast, shorter when slow

```
desired_linear_vel: 0.6       # well under the 1.4 m/s cap
lookahead_dist: 0.8
min_lookahead_dist: 0.4
max_lookahead_dist: 1.2
use_velocity_scaled_lookahead_dist: true
min_turning_radius: 1.6
```

Two parameters that cannot both be true

```
use_rotate_to_heading: false
allow_reversing: true
```

Set both `true` and **Nav2 refuses to start.**

Rotate-to-heading means **"stop and spin to face the right way"** —
which this vehicle cannot do at all.

So `false` is not a preference. It is the only legal value for a car —
and it is what makes `allow_reversing` available.

3. The constraint reaches the goal checker

These look sloppy. They are geometry.

```
goal_checker:
  xy_goal_tolerance: 0.6
  yaw_goal_tolerance: 0.6
```

`min_turning_radius: 1.6` = R_min 1.52 m + margin.

A car with that radius and no in-place rotation **cannot reduce a 0.75 m pose error.** It can only loop past and retry.

0.6 m is ~40% of the turning radius — achievable in one pass.

F-N3: the car that orbited forever

At 0.25 m tolerance the vehicle arrived within 0.75 m, then circled:

```
(-4.19,-1.46) (-5.41, 0.43) (-2.28, 0.81) (-4.51,-1.35) ...
```

It never converged. It never failed.

No error. No timeout. No diagnostic.

Just a car driving in circles around a point it could not reach.

The system did not report a problem, because from its own point of view there was not one.

Weeks 1, 3, 4 — and now 9. Every component did exactly what it was configured to do.

Track work

Track	
Electrical	Close the loop on the bench. **≤ 1°, no oscillation**
Chassis	Measured `mitt_dimensions.yaml` committed — two tracks are waiting
Simulation	`mitt_bench` skeleton: measure something, print a number
Software	`mitt_hardware` read()/write() against a recorded bag

Merge 1 is 11/23 — two weeks.

If either half will not be ready, say so at **today's** standup.